

[WORKING TITLE: Paid Access: Test-Time Compute on a Conservative Memory as a Physically-Metered Resource]

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Abstract

Test-time compute — retries, escalation, non-local routing — is usually a black box bolted onto a trained model: more forward passes, a learned halting head, no account of what capability the extra compute buys or what it costs the model’s guarantees. We take a **position**: when the underlying memory is a *conservative* (symplectic) associative memory, every mechanism that buys capability at inference can be made to carry an explicit **physical receipt** — a phase-volume Jacobian, a bounded energy ledger, an exact latch-transport law — and the resulting statements are **theorems about the mechanism, not empirical scaling curves**. We develop this “paid access” frame for a single unit whose latent state (q, p) is advanced by a damped symplectic (velocity-Verlet) step of a learned Hamiltonian, and split reachability into two provably distinct failure modes: **reach** (the target lies outside a kinematic *causal box* C_T , half-widths $L_i = T\epsilon c/\sqrt{M_i}$, set by velocity not energy) and **escape** (inside the box, behind a barrier). A **Lorentz squeeze** cures escape with a bounded, governor-re-absorbable injection $\leq e^{2|\zeta|}H$ and *prices reach rather than capping it* (its instantaneous displacement $p_0 \sinh \zeta/M_0$ carries the state beyond the energy-blind box, but at an energy cost growing exponentially in rapidity); an **intra-unit wormhole** (a gated canonical translation) cures reach by teleporting across the box at $\det J = 1$ *exactly*, paying only a *fixed* discrete ledger $\Delta V = V(b) - V(a)$ independent of distance and **transporting** any latched content by an exact $p^\top X \Delta$ — the load-bearing dichotomy as a **pricing law**: squeeze reach is exponentially priced in distance, wormhole reach is flat-priced. On a designed analytic testbed (dim 2 and 4, 5 seeds, oracle placement, laptop-CPU) we *verify* the full certificate stack — squeeze reach steps up and is then priced out past the swept rapidity budget $\zeta \leq 2.0$ (its reach is the bracket $[L, L + p_0 \sinh \zeta/M_0]$, not a knife-edge; the same bracket predicts $\zeta \approx 2.01$ to reach $d = 4.0$), the wormhole lands flat at all distances with $\det J = 1$ and ledger = 0 exact, and a Newtonian-mode control reaches past the box (confirming the cap is the constraint). The receipt is not a label: on this designed testbed it separates **transport** from **state-replacement**. The wormhole’s canonical translation *transports* a stored Goldstone charge by the exact $p^\top X \Delta$ (spread preserved, $\text{std}(Q_{\text{out}}) = \text{std}(Q_{\text{in}}) = 0.0803$); an untrained **state-replacing** map $(q, p) \mapsto (b, p)$ — a no-physics baseline that reaches by overwriting the state — has $\det J = 0$ *exactly* (measured) and **erases** it ($\text{std}(Q_{\text{out}}) = 0$). Deciding *whether* to take a certified edge is orthogonal: a learned decision head (§4.2) routes *through* the $\det J = 1$ channel and inherits its receipt — the certificate prices only transport. The fine print travels with the claim: volume preservation *alone* is not the latch receipt (a $\det J = 1$ random shift also scrambles the charge — the *matched channel* is what preserves it), and a bounded — even *free* — energy ledger does not buy BIBO (a $\Delta H = 0$ exit into a non-coercive region escapes with $r^* \propto T$; coercive-*component* membership is the operative clause). We then situate three *honest* pillars on **learned** memories, each scoped and priced in-line: a distribution-free calibrated compute-rationing gate whose mechanism is **memory-agnostic**; a direct one-hop non-local edge whose cost is flat in the number of units where multi-hop diffusion scales; and a settled

regime map on which a modern Hopfield network is the cheaper *and* more cue-noise-robust retriever at matched accuracy, the CLU gate reaching Hopfield accuracy only for clean/correlated cues at small capacity. The contribution is the **certificate stack and its discipline**, not a benchmark win.

1 Introduction

Adaptive test-time compute is a standard lever: models spend more inference budget on harder inputs via learned halting (ACT, Graves 2016; PonderNet, Banino et al. 2021), confidence-gated early exit (CALM, Schuster et al. 2022), learned routing (Mixture-of-Depths, Raposo et al. 2024; Mixture-of-Experts, Shazeer et al. 2017), and energy-as-verifier “thinking” (Energy-Based Transformers, Gladstone et al. 2025). Across this literature the gate is a *learned scalar* bolted onto a feedforward stack, and “more compute” means *more of the same operation* with no structural guarantee on what that operation does to the model’s state.

This paper takes a different starting point and a **position**. When the memory being queried is a *conservative* dynamical system — a latent state (q, p) advanced by a **structure-preserving symplectic integrator** of a learned separable Hamiltonian $H(q, p) = T(p) + V_\theta(q)$, with optional per-step damping supplying controllable forgetting — then test-time compute is **access to a phase space**, and the physics of that phase space *prices every access mechanism explicitly*. We call this **paid access**: each mechanism carries a **physical receipt** — the Jacobian determinant $\det J$ (did it conserve phase volume?), a bounded energy ledger (how much did it inject, and where does that energy go?), and a latch-transport law (what did it do to content stored in a protected direction?). The receipts are **certificates that follow from the symplectic structure, not curves fit to runs**. Our thesis is that this is the right way to reason about test-time compute on a conservative memory, and it is falsifiable: we state each certificate as a theorem and verify it, to machine precision, on a designed testbed.

The reference unit. Our reference memory is the **CLU (Causal Learning Unit)**, introduced as **CHLU in Jawahar & Pierini (2026)**; the exactly-solvable theory whose certificates we verify is developed in a companion note (Anonymous, 2026; hereafter *the theory note*). Nothing below uses a property specific to the reference training objective except where stated. **Nomenclature (do not conflate)**: the learned **inertial mass** M_i (diagonal of T ; larger $M \Rightarrow$ slower, smaller light-cone) and the **spectral mass** μ_k (a normal-mode stiffness) run in opposite directions. Reach statements use M ; retention statements use μ .

Contributions.

1. **The paid-access frame and its two failure modes (§2–3).** Reachability splits into *reach* (kinematic; target outside the causal box C_T , $L_i = T\varepsilon c/\sqrt{M_i}$) and *escape* (energetic; inside, behind a barrier); the two are cured by different mechanisms with different receipts. *[proven.]*
2. **The certificate stack, verified (§3, headline).** On a designed testbed (dim 2/4, 5 seeds, oracle placement, laptop-CPU) the squeeze cures escape with bounded injection $\leq e^{2|\zeta|}H$ and *prices reach* (its reach is the bracket $[L, L + p_0 \sinh \zeta/M_0]$ — it exceeds C_T only by its instantaneous displacement, at energy growing exponentially in ζ); the wormhole cures reach at $\det J = 1$, a *fixed* ledger = 0 exact, transporting the latch by exact $p^\top X\Delta$; a dense potential fails reach; a Newtonian control confirms the cap. *[verification of the theory’s exactness; oracle placement.]*

3. **The receipt has a measured downstream consequence (§3.2.1, headline panel).** The certificate is not a decorative label. The router’s map $(q, p) \mapsto (b, p)$ has $\det J = 0$ *exactly* (measured by autodiff, not asserted) — volume-annihilating and non-invertible — so it collapses a capture ball of stored states onto a single charge, **erasing** the Goldstone spread ($\text{std}(Q_{\text{out}}) = 0$, all 16 probes bit-identical), where the wormhole **transports** it ($\text{std}(Q_{\text{out}}) = \text{std}(Q_{\text{in}}) = 0.0803$; $\Delta Q = p^\top X \Delta$ to 1.2×10^{-7} ; reconstruction exact to 2.2×10^{-8}). **Fine print travels with the claim:** volume alone is *not* the latch receipt (a $\det J = 1$ random shift also scrambles the charge), and a bounded — even free ($\Delta H = 0$) — ledger is *not* sufficient for BIBO (§3.1, App. B.2). [*verification; designed testbed, oracle placement.*]
4. **Calibrated compute-rationing on an escalatable learned memory (§4.1).** A self-calibrating gate rations relaxation budget on a *trained* memory ($4.81 \pm 0.44 \times$ fewer relaxation steps **at kv16, at a 400-epoch budget**; the payoff decays to $1.57 \times$ at kv24 and $1.14 \times$ at kv32; never below always-full at any level; 5 seeds, MQAR vocab-256, laptop), with Learn-then-Test coverage certificates (30/30 valid). The *saving* is the robust invariant; the *accuracy gain* over always-full is a 400-ep (not-yet-converged) property — at convergence (§4.3, App. C.4.a) the gate’s accuracy **matches** full-budget, and the payoff is rationing, not accuracy. The gate mechanism is **memory-agnostic**; the CLU-conditional asset is *escalatability*, not a superior energy signal. [*evidence.*]
5. **The one-hop non-local edge, and its boundary (§4.2).** A direct wormhole edge has cost flat in the number of units where multi-hop diffusion scales (1.18×10^8 vs $1.76 \rightarrow 2.94 \times 10^8$ FLOP/query, $N \in \{4, 8\}$) with better distant accuracy — but *energy-gating* it **loses** to a 449-param physics-free router in FLOPs and accuracy, reported plainly as the boundary. [*evidence.*]
6. **The regime map, settled, and the corrected cost story (§4.3).** At matched accuracy a modern Hopfield network is the **cheaper and more cue-noise-robust** retriever (≈ 1 matvec); the CLU gate’s 9–10 $\times$ figure is *intra-CLU* rationing vs a full-budget CLU, never a cost win over Hopfield. On the full grid (198 jobs, $n = 8$) the gate reaches/reverses Hopfield **only on clean/correlated cues at kv ≤ 64 ($\Delta + 0.02$); beyond kv64 is an epoch-budget wall, not a capacity wall**; and — the dominant negative — **no cell closes under cue noise** (gate 0.36 vs Hopfield 0.71 at $\sigma = 0.6/\text{kv}32$ despite fidelity ≈ 1.0). The gate rations *clean* retrieval; noise-robustness is Hopfield’s. [*evidence; cost final, accuracy regime-specific and settled.*]

The paper’s value is the certificate stack of §3 and the discipline of §4, not a leaderboard result. Everything is laptop-CPU. **We own the shape of this contribution profile explicitly:** the certificate stack is the contribution, and it is verified where we control the testbed; §4 maps precisely where it does and does not translate into a learned-memory advantage over a cheaper black box, and three of our six contributions are boundaries or negatives. §4 is the paper’s honest perimeter, not an incidental ablation.

2 Setup: a conservative memory and its reachable set

The map. One dissipative velocity-Verlet step with step ε and per-step damping $\gamma \in [0, 1)$ advances (q, p) by a kick–drift–kick of $H = T(p) + V_\theta(q)$ then $p \mapsto (1 - \gamma)p$. The update is **conformally symplectic**: $J^\top \Omega J = (1 - \gamma)\Omega$, $\det J = (1 - \gamma)^d$ in dimension d ; a position-gated coupling scales this to $\det J = (1 - \gamma\varphi(q'))^d$. A **kinetic mode** fixes T : *relativistic* mode hard-caps $|\dot{q}_i| < c/\sqrt{M_i}$; *Newtonian* mode has $\dot{q}_i = p_i/M_i$ unbounded. A state-dependent **governor**

$\gamma_n = s \tanh(\max(0, H - E^*))$ bleeds excess energy toward E^* . Full derivations and machine-precision checks are in the theory note.

Reachability, made falsifiable. For the governed map $\Phi_{\varepsilon, \gamma}$, the T -step **reachable set** from z_0 is $R_T(z_0) = \{\Phi^n(z_0) : 1 \leq n \leq T\}$, position shadow $Q_T(z_0)$. **Access** is provably enlarging Q_T . In relativistic mode one drift advances q by at most $\varepsilon c / \sqrt{M_i}$ per coordinate, so

$$Q_T(z_0) \subseteq C_T(q_0) := \{q : |q_i - q_{0,i}| \leq L_i, L_i = T\varepsilon c / \sqrt{M_i}\}.$$

The **causal box** C_T is **energy-blind**: injected momentum drives $|\dot{q}_i|$ toward but never past $c / \sqrt{M_i}$. Hence the dichotomy:

- **REACH failure:** $q^* \notin C_T(q_0)$ (kinematic). Curable *only* by a nonlocal jump.
- **ESCAPE failure:** $q^* \in C_T(q_0)$ but trapped behind a barrier ΔV_b the dissipating budget cannot climb. Curable by any bounded injection faster than the governor re-brakes.

In Newtonian mode reach is instead energy-limited ($|\Delta q_i|_T \leq T\varepsilon \sqrt{2(H - V_{\min})/M_i}$), so energy *does* buy reach — which is why the safe headline mode (relativistic) is the mode where energy *cannot* buy *flow* reach (the cap bounds the Verlet drift regardless of budget), motivating a nonlocal mechanism. A squeeze can still buy reach *canonically* — as an instantaneous displacement, not through the flow — but only at the exponential energy price §3 makes explicit; that is the pricing law, not a loophole in the cap. The two failure modes need *separate* predictions (the disambiguation §3.3 returns to).

3 The certificate stack, verified

Reporting grade: verification. Every number in §3 is on a designed, architecturally-invariant analytic testbed (a symmetric double well along the reach coordinate; an $SO(2)$ sector for the latch), with **oracle channel placement**, dim 2 (headline) and 4 (identical), 5 seeds, laptop-CPU, $\gamma = 0$ rollout for a sharp box, mass band $[4.0, 0.25]$ ($16\times$ inertial contrast). These verify the theory's *exactness*; learned entrance-placement is out of scope (§5).

3.1 The two mechanisms and their receipts

Squeeze cures escape (bounded, re-absorbable injection). A mass-weighted Lorentz squeeze $S_\zeta^{(M)}$ is symplectic ($\det = 1$ exactly) and delivers both amplified velocity and displacement toward the saddle, $\delta q' = \delta q \cosh \zeta + p_0 \sinh \zeta$. Its injected energy is **bounded**, $H(S_\zeta z) \leq e^{2|\zeta|} H(z)$ (theory note Prop-12); the governor re-absorbs it in $\approx 2\zeta/\gamma_c$ steps. *Verified:* on the matched quadratic H the injection tracks the bound at every rapidity — $(\zeta, \text{ratio}, \text{bound}) = (0.25, 1.13, 1.65), (0.5, 1.55, 2.72), (1.0, 3.79, 7.39), (2.0, 27.5, 54.6)$ — and $\det S^{(M)} = 1.000 (\pm 4 \times 10^{-6})$. **Fine print (stated with the claim):** the $e^{2|\zeta|}$ bound is a *matched-quadratic- H* certificate; on the quartic well the raw ratio can exceed it (expected), so we quote it in that scope.

Wormhole cures reach ($\det J = 1$ exact, ledgered, latch transported). An intra-unit wormhole is a learned pair of loci (a, b) with a hard gate frozen at capture, applying the **constant canonical translation** $q \mapsto q + \Delta$, $p \mapsto p$, $\Delta = b - a$. Its Jacobian is I_{2d} , so $\det J = 1$ **exactly**, **independent of Δ and the gate** (the flow of the linear Hamiltonian $W = \Delta \cdot p/\tau$). Its only

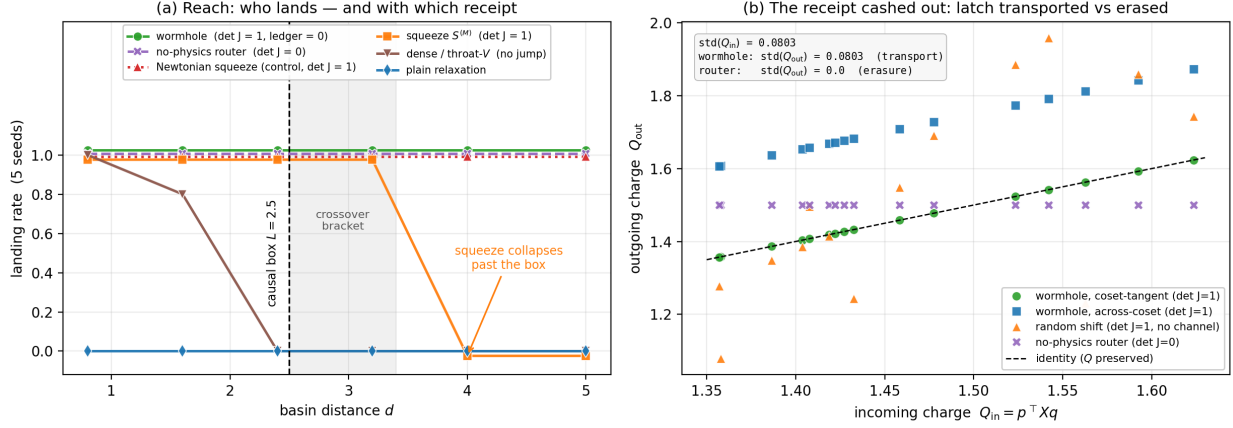


Figure 1: **Headline (§3, verification grade).** (a) **Reach:** landing rate vs. basin distance d on the designed double-well testbed (dim 2, 5 seeds, oracle placement, $L = 2.5$; *arms are offset vertically for visibility* — wormhole, state-replacing map and Newtonian control all land at exactly 1.0). Plain relaxation is escape-blocked everywhere; the squeeze $S^{(M)}$ steps up and its reach is then priced out past the swept rapidity budget $\zeta \leq 2.0$ (its reach is the bracket $[L, L + p_0 \sinh \zeta / M_0]$, shaded — the observed edge is $d \approx 3.2$, not a knife-edge at L ; the same bracket predicts $\zeta \approx 2.01$ to reach $d = 4.0$ and $\zeta \approx 2.64$ for $d = 5.0$, at exponentially rising energy — reach is *priced*, not capped); the wormhole is flat at all d with $\det J = 1$ and a *fixed* ledger = 0 exact; the Newtonian-squeeze control reaches past L , confirming the relativistic cap is the operative constraint on the flow; the dense/throat- V potential fails reach for $d \geq 2.4$. (b) **The receipt cashed out:** outgoing vs. incoming Goldstone charge for 16 states drawn in the capture ball. The $\det J = 1$ wormhole **transports** the charge (slope 1; exact constant shift $p^\top X\Delta$; spread preserved, $\text{std}(Q_{out}) = \text{std}(Q_{in}) = 0.0803$) while the untrained $\det J = 0$ **state-replacing** map **erases** it (slope 0; all 16 states exit at one common charge, $\text{std}(Q_{out}) = 0$). The fine print is in the figure too: a $\det J = 1$ *random shift* also scrambles Q — volume alone is not the latch receipt; the matched channel is.

cost is a discrete jump $\Delta H_{wh} = V_\theta(q + \Delta) - V_\theta(q)$ that must be **ledgered**; matched loci give free transport. *Verified:* symmetric double well, channel $-a \rightarrow +a$, ledger = 0.0 exact, state teleports to $q = +1.000$; $\det J = 1.000$. **Fine print:** a gate varying *during* the jump gives $\det J = 1 + \nabla g \cdot \Delta \neq 1$ (unit test 2.05) — an unpaid contraction; the receipt is clean only for a frozen gate (a proven design guard, App. D N31).

Latch transport ($p^\top X\Delta$, exact). For a protected direction carrying Goldstone charge $Q = p^\top Xq$, under the wormhole $Q' - Q = p^\top X\Delta$ *exactly*. The wormhole moves *one* phase point, so it **transports** the latch (not copy, not erase): preserved iff $\Delta \perp X^\top p$, else shifted by the exact $p^\top X\Delta$. *Verified:* zero-shift $\Delta Q = 0.0$ (pred. 7.5×10^{-10}); across-coset $\Delta Q = 0.2500 = p^\top X\Delta$; raw squeeze preserves Q ($\Delta Q = 1.2 \times 10^{-7}$); random shift erases ($\Delta Q \in \{0.035, 0.157, -0.143, 0.302, -0.144\}$). The receipt distinguishes *transport* from *erase* — the certificate that a paid jump carries memory. **Fine print, stated with the claim (C-6):** $\det J = 1$ *alone* is not the latch receipt — the random-shift baseline is itself volume-preserving and still scrambles Q ; what buys transport is the **matched channel** (Δ coset-tangent, $X\Delta \perp p$) together with the ledger.

BIBO fine print, stated with the wormhole claim (C-6). $\det J = 1$ certifies *volume*, not *boundedness*. An exit placed outside the coercive connected component of V_θ can escape to infinity

even though the jump is symplectic and its ledger bounded — and, sharper, *even when the ledger is free*. On a controlled non-coercive well ($V = \frac{1}{2}kq_0^2 - \epsilon q_0^4$, coercive edge $x_b = 3.536$) the exit at $b = 5.0$ has $\Delta H = 0.0$ *exactly* — cheaper than the admissible $b = 3.0$ exit’s $\Delta H = 2.88$ — and an energy-only sub-level test **admits** it, yet the trajectory escapes with $r^* \propto T$ (growth $r^*(2T)/r^*(T) = 1.91$), while the screened exit stays bounded (ratio 1.000). **Coercive-component membership, not the energy ledger, is the operative BIBO clause** (6/6 exits predicted; App. B.2, Fig. 4). Full receipt: Table 1 (App. B).

3.2 The discriminating experiment: reach steps then collapses; the wormhole is flat — and only it hands back a receipt

The falsifiable heart is a single controlled reach battery: a K -basin double well with basins spanning below/above the box $L = 2.5$ ($T = 100$, $\varepsilon = 0.05$, $c = 1$, heavy reach coord $M_0 = 4.0$), the governor fixed so plain relaxation *provably* cannot leave the start basin (escape-blocked, $\text{KE}_0 = 0.72 < \Delta V_b = 1$). Figure 1 is the headline: panel (a) the landing rates, panel (b) the receipt that separates the arms landing rates cannot. Landing rate vs. distance d (dim 2, 5 seeds; $< L = \{0.8, 1.6, 2.4\}$, $> L = \{3.2, 4.0, 5.0\}$; dim 4 identical):

arm	$d < L$	$d > L$	det J (measured)	reading
plain relaxation	0	0	$(1-\gamma)^d$	escape-blocked everywhere
squeeze $S^{(M)}$	1	$\rightarrow 0^\dagger$	$1.000 \pm 4\text{e-}6$	steps up; reach beyond the box is priced
wormhole	1	1	1.0 exact, ledger 0.0	flat ≈ 1 at all d , with a receipt
Newtonian-squeeze (control)	1	1	1.0	energy buys flow reach — confirms the cap
state-replacing map (no-physics)	1	1	0.0 exact	reaches by <i>overwriting</i> , volume-annihilating
dense/throat- V	$\rightarrow 0$	0	$(1-\gamma)^d$	helps near, fails reach

[†] The squeeze’s $\rightarrow 0$ at $d \in \{4.0, 5.0\}$ means “priced out of the swept $\zeta \leq 2.0$ grid,” *not* “cannot reach”: the verified bracket predicts $d = 4.0$ lands at $\zeta \geq 2.0105$ ($\approx e^{4.02}H$) and $d = 5.0$ at $\zeta \geq 2.6441$ ($\approx e^{5.29}H$) — analytic predictions from the bracket, not new runs; the grid ended at $\zeta = 2.0$. **Two “router” objects:** the `no_physics_router` arm here is an *untrained analytic constant map* (a state-replacing baseline), distinct from §4.2’s *learned decision head*, which routes *through* the wormhole’s det $J = 1$ edge and inherits its receipt.

Figure 1 is deliberately two-panel because the landing-rate axis *cannot* display this paper’s thesis: the wormhole, the state-replacing map and the Newtonian control all land at 1.0 and coincide. What distinguishes them is the receipt column — and panel (b) is that column made visible.

What each arm certifies. (i) The **squeeze prices reach; it does not fail at the box**. It is a canonical map, not a flow: applied at the basin bottom it grants an *instantaneous* displacement $(p_0/M_0) \sinh \zeta$ before the capped flow adds at most L , so its reach is the bracket $[L, L + p_0 \sinh \zeta / M_0]$ — it *does* exceed the box, by a ζ -controlled amount (observed edge $d \approx 3.2$, not a knife-edge at $L = 2.5$). But that excess costs energy $\leq e^{2|\zeta|}H$ (§3.1), so reach beyond L carries an **exponential price in rapidity**: the swept $\zeta \leq 2.0$ reaches $d \lesssim 3.6$ ($\leq e^4 \approx 55H$), and the same verified bracket predicts $d = 4.0$ needs $\zeta \approx 2.01$, $d = 5.0$ needs $\zeta \approx 2.64$. The falsifiable content, **sharpened from a collapse into a pricing law**: *reach via squeeze is exponentially priced in distance; reach via wormhole is flat-priced*. The squeeze’s $\rightarrow 0$ entries mean “priced out of the swept grid,” not “cannot reach.” (ii) The **wormhole** is flat at all d with det $J = 1$ and a *fixed* ledger = 0 per jump, independent of d . (iii) The **Newtonian control** reaches past L (energy buys *flow* reach uncapped) — the control showing the relativistic cap, not a coding artifact, is the constraint on the flow. (iv) The **state-replacing map** (the untrained no-physics baseline) matches the wormhole’s landing but by fiat, by *overwriting* the state. Its map $(q, p) \mapsto (b, p)$ has Jacobian $\text{blockdiag}(0_d, I_d)$, so det $J = 0$

exactly (measured by forward-mode autodiff): *volume-annihilating and non-invertible*, a strictly stronger statement than “carries no certificate,” cashed out in §3.2.1. **It is not §4.2’s learned router**, a decision head that routes *through* the wormhole’s $\det J = 1$ edge (decision and transport are orthogonal). (v) The **dense/throat- V** arm fails reach for $d \geq 2.4$: a nonlocal potential lowers the barrier but the trajectory still traverses it, staying bound by C_T . The wormhole is the *only* arm reaching all d **at a fixed ledger with a $\det J = 1$ receipt**.

3.2.1 The receipt cashed out: a state-replacing jump erases the latch a canonical translation transports

A certificate that never changes an outcome is decoration. Here it changes one, measured on the same designed testbed (Fig. 1b; dim 2 headline, replicated at dim 4).

Decision is not transport — state this once, sharply. Two objects in this paper are loosely called “router,” with opposite transport semantics. §4.2’s **router_mlp** is a *learned decision head*: it *decides whether* a query takes the non-local edge and transports *through* the wormhole’s own $\det J = 1$ channel (“routes via the **same** direct wormhole edge”), inheriting the receipt and erasing nothing. The object here is different in kind — an *untrained analytic map* that *is* the transport, and a **state-replacing** one: it overwrites $(q, p) \mapsto (b, p)$, annihilating phase volume. **A learned gate bolted onto a certified channel is not a counterexample to the receipt; it is a consumer of it.** The certificate prices only transport, never the decision.

We draw 16 incoming states uniformly in the capture ball (radius $0.3 < \rho = 0.35$) around an entrance on the vacuum circle at fixed momentum p , and read $Q = p^\top X q$ for each; the incoming cloud has $\text{std}(Q_{\text{in}}) = 0.0803$.

arm	$\det J$	ΔQ mean	ΔQ std	pred. $p^\top X \Delta$	max err	$\text{std}(Q_{\text{out}})$
wormhole, coset-tangent	1.0000	−0.0000	0.0000	0.0000	$1.2\text{e}−7$	0.0803
wormhole, across-coset	1.0000	0.2500	0.0000	0.2500	0.0	0.0803
random shift (no channel)	1.0000	0.0972	0.2465	(none)	—	0.2793
state-replacing map	0.0000	0.0379	0.0803	(none)	—	0.0

The guarantee. The wormhole’s canonical translation is *injective*, so it shifts **every** incoming state’s charge by the **same exact constant** $p^\top X \Delta$ (ΔQ std = 0, err $\leq 1.2 \times 10^{-7}$). The stored spread survives: $\text{std}(Q_{\text{out}}) = \text{std}(Q_{\text{in}})$ to all printed digits, and the pre-jump state is exactly recoverable, $q_{\text{in}} = q_{\text{out}} - \Delta$ (max err 2.2×10^{-8}).

The violation. The state-replacing map sends the whole capture ball onto one point: all 16 states exit at a single common charge (max $|Q_{\text{out}} - Q_{\text{out}}[0]| = 0.0$, bit-identical) and $\text{std}(Q_{\text{out}}) = 0$ **exactly** — the latched coset content is *irrecoverable*. This is $\det J = 0$ cashed out as a measured downstream consequence, replicating in every cell ($\text{dim} \in \{2, 4\} \times \text{seed} \in \{0, 7\}$: $\text{std}(Q_{\text{out}})_{\text{wh}} = \text{std}(Q_{\text{in}})$ in all four — 0.0803/0.0679/0.0533/0.0448 — and $\text{std}(Q_{\text{out}})_{\text{map}} = 0$ in all four).

The claim at its honest altitude. The receipt separates **transport** from **state-replacement**, with a measured consequence: at $\det J = 1$ the matched canonical channel *transports* the stored charge; a $\det J = 0$ state-replacing jump *erases* it. The claim is about *maps*, not ideologies or learnedness — which makes it robust. Three qualifications travel with that sentence. (i) **Volume alone is not the latch receipt** — the random-shift arm has $\det J = 1$ and still scrambles Q (out-spread 0.2793, $3.5 \times$ the incoming spread); the *matched channel* preserves it. (ii) This is a **mechanism-level violation on a designed testbed with oracle placement**, not a learned-system win, and the erased object is the *untrained state-replacing baseline*, not §4.2’s learned router. §4.2’s **router_mlp** is a **decision head**: it beats the energy-gated edge in FLOPs

and accuracy at *choosing* whether to take the non-local edge, by transporting *through* the wormhole’s own $\det J = 1$ channel — inheriting the receipt, erasing nothing. Decision and transport are orthogonal; both facts hold. (iii) For the BIBO half (App. B.2) the attribution is narrower: what buys boundedness is **the receipt, not the jump**, since an unscreened wormhole and the state-replacing map coincide exactly.

3.3 Why a prior null does not bear on this claim

A memory of this class was previously reported to gain nothing from squeeze/boost *retrieves* at single-unit scale (a pooled null; App. D N1). That null tested **selection among already-reachable** attractors (retrieving the wrong pattern) at near-uniform mass ($S^{(M)} \approx S$) with within-basin perturbations — none of its targets sat *outside* the reachable set. The reach claim asks a categorically different question — land in a basin *provably outside* plain relaxation’s reach, with a $16\times$ mass contrast — verified by the §3.2 crossover. We flag this to preempt “didn’t your own retrievers fail?”: the two share only the operator $S^{(M)}$.

4 Three honest pillars on learned memories

Reporting grade: evidence. §4 leaves the designed testbed for **trained** memories; each claim is scoped and priced in-line. Where the receipt says a cheaper black box wins, we say so.

4.1 Calibrated compute-rationing — memory-agnostic mechanism, escalatable-memory asset

We equip a trained CLU associative memory (MQAR key-value recall, vocab 256, $\text{kv} \in \{16, 24, 32\}$, 5 seeds, laptop) with a self-calibrating gate: at write time it runs a jittered-cue self-test and fits a per-model calibration head (Platt 1999) mapping a relaxation residual to p_{wrong} , plus a learned exit threshold by **Learn-then-Test** (LTT; Angelopoulos et al. 2021). Deployment queries run a staged-relaxation ladder and exit when calibrated-confident. Three findings:

1. **Calibration transfers, and the mechanism is memory-agnostic.** Raw residual energy is not cross-model comparable (pooled AUROC 0.431 ± 0.038 , anti-ranked); the per-model head makes it deployable (0.869 ± 0.015 , 5 seeds). The *identical* stack on a modern Hopfield memory (Ramsauer et al. 2021) reproduces the jump (raw $0.18 \rightarrow$ calibrated 0.88 , matching CLU $0.43 \rightarrow 0.87$) — the gate mechanism is **not** CLU-specific. The value is the calibrated-rationing apparatus, not a claim that CLU energy is a better confidence signal.
2. **The CLU-conditional asset is escalatability — and the allocation payoff is largest at the easy band.** What does not transfer to a one-shot memory is the allocation payoff: on the CLU’s graded relaxation the learned point spends $4.81 \pm 0.44\times$ fewer steps **at kv16, at a 400-epoch budget** (0.894 ± 0.021 @ 629 ± 60 vs always-full 0.847 ± 0.037 @ 3000) while landing *above* always-full accuracy. **The epoch budget is load-bearing and we own it inline (C-7):** §4.1’s models are trained 400 ep, a band §4.3 shows is not yet converged. At convergence (§4.3, App. C.4.a: 2000 ep) the gate’s accuracy **matches** full-budget rather than exceeding it — the robust payoff is *rationing* ($9.9\times$ intra-CLU), and the accuracy headroom is a property of an imperfect (under-trained) memory, not of the mechanism; CM-2’s “escalatable memory” claim rests on the saving. **The magnitude of the saving also does not persist as the band hardens, and we name the levels rather than the band:** kv24 gives $1.57 \pm 0.07\times$ (0.547 ± 0.039 @ 1919 ± 82 , +2.2 pts over always-full) and kv32 gives $1.14 \pm 0.06\times$ (0.286 ± 0.037

@ 2636 ± 127 , +0.6 pts). What survives at all three levels is the weaker, cleaner invariant: **the learned gate never pays full price for less accuracy at any level**, and savings scale with how much of the workload is confidently easy — a $4.8\times$ headline is a kv-16 number, not a property of the mechanism. A one-shot memory earns no such payoff at *any* level. Scoped to MQAR vocab-256, kv $\in \{16, 24, 32\}$, 5 seeds, laptop.

- 3. Distribution-free certificates, assumption stated.** The LTT wrapper was valid on 30/30 cells (zero violations), e.g. coverage 0.647 ± 0.063 at target risk $\varepsilon = 0.05$ (measured risk 0.030), with graceful refusal where the memory is too weak. **Fine print, next to the claim:** LTT rests on an *exchangeability* assumption between write-time probes and deployment queries, which our jittered protocol only approximates — the calibration is *under-confident* (ECE $\approx 0.100 \pm 0.021$, safe side but a real shift), so the guarantee holds “within the stated probe-to-deployment scope.”

Two negatives on the record (App. D). (a) The gate does not let this memory out-abstain a near-perfect Hopfield (0.983–1.000 accuracy $\Rightarrow$ selective-risk ≈ 0 ; N2). (b) Residual energy adds essentially nothing over the readout margin ($\Delta\text{AUROC} \in [-0.004, +0.024]$); energy-as-a-superior-signal is a claim we **do not make** anywhere (N3). The pillar is the certified rationing apparatus on an escalatable memory.

4.2 The one-hop non-local edge, and where energy-gating it loses

A conservative memory can be given a **non-local edge**: a gated wormhole coupling transporting a query key to a distant archive, an attention-like long-range access. The surviving mechanism claim is about *cost structure*: the **direct one-hop edge has cost flat in N** (1.18×10^8 FLOP/query at $N \in \{4, 8\}$) where **N -hop chain diffusion scales** ($1.76 \times 10^8 \rightarrow 2.94 \times 10^8$) *and* loses distant accuracy ($0.41 \rightarrow 0.28$) — a direct edge beats hop-by-hop diffusion, in FLOPs and accuracy (5 seeds, MQAR-style, $N \leq 8$, laptop).

The honest boundary, stated plainly as the section’s spine: **energy-gating that edge loses**. A parameter-matched (449-param, 2-layer) physics-free router on the raw cue beats the energy-gated wormhole in *both* FLOPs and accuracy — router 1.000/0.948 (local/distant, $N = 4/8$) @ 8.81×10^7 FLOP vs gated 0.887/0.715 @ 1.18×10^8 — across all mixes $\{50/50, 80/20, 95/5\}$ and both N , over 5 seeds. Near-linearly-separable per-unit key clusters let the cheap classifier learn local-vs-distant trivially while the energy gate rediscovers the partition via relaxation, noisier. **The direct edge is the mechanism; energy is not the routing signal.** (Scope caveat, inline: linearly-separable cues drive the router’s dominance; a harder band is the untested fair stress test; App. D N24/N27.)

4.3 The regime map and the corrected cost story

Cost (final). A modern Hopfield network reaches its ceiling in ≈ 1 matvec (0.947–0.979 at $\beta \geq 5$; extra iterations change accuracy ≤ 0.003), so its floor is $O(\text{kv} \cdot d)$. **At matched accuracy, Hopfield is the cheaper retriever.** The CLU gate’s “9–10 $\times$ savings” is an **intra-CLU** number — gate vs full-budget CLU (3000 Verlet steps) — a compute-rationing measurement, **not** a cost win over Hopfield. “Matching Hopfield” means the gate’s accuracy *reaches* Hopfield’s, not that it does so more cheaply. We state this next to every accuracy-improvement claim.

Accuracy (settled — full grid, 198 jobs). The initial “Hopfield-dominant 26/26” map was an **under-training artifact** (500 ep: gate 0.02–0.31 vs Hopfield 0.95–0.98, confirmed $n = 8$). Trained to convergence (2000 ep) CLU fidelity rises to ≈ 1.0 and the accuracy reversal appears **but is**

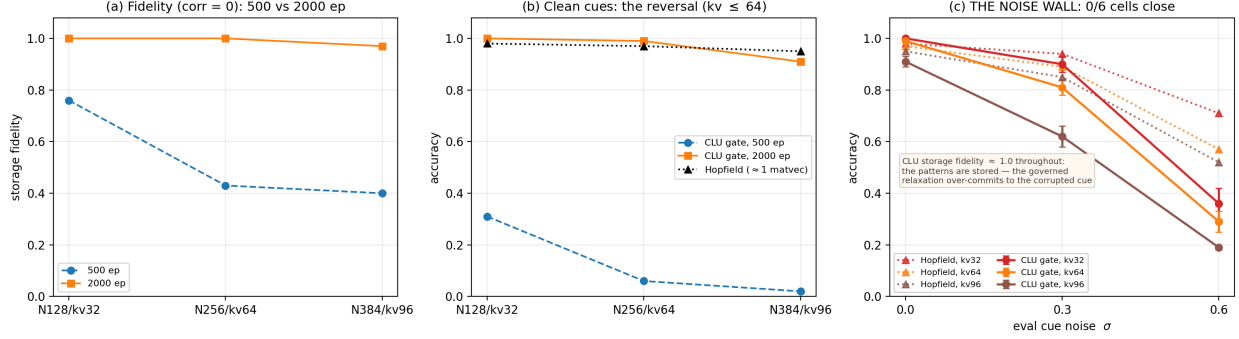


Figure 2: **The settled regime map (§4.3, evidence; regime-remap-2000ep, 198 jobs).** (a) Storage fidelity on the paired capacity axis, 500 vs 2000 ep ($n=8$): the 500-ep “Hopfield-dominant” map was an under-training artifact. (b) Clean cues: at 2000 ep the gate reaches/reverses Hopfield for $kv \leq 64$ ($\Delta + 0.02$) at $9\text{--}10\times$ *intra-CLU* rationing (never a cost win over Hopfield, which answers in ≈ 1 matvec). (c) **THE NOISE WALL — the dominant negative, plotted.** Under eval cue noise $\sigma \in \{0.3, 0.6\}$ the gate (solid) falls below Hopfield (dotted) at every capacity — 0/6 cells close — *despite* CLU storage fidelity ≈ 1.0 : the patterns are stored, but the governed relaxation over-commits to the corrupted cue. Error bars are ± 1 s.d. over seeds for the gate ($n=5$ at $\sigma > 0$; $n=8$ at the $\sigma=0$ clean reference, taken from the $\text{corr}=0$ capacity axis); the source reports no per-seed spread for the stress-axis Hopfield arm, so its curves carry no bars.

regime-specific, not general, and never touches the cost story above (every “9–10 $\times$ ” here is *intra-CLU* rationing, not a win over Hopfield). Three qualifiers travel together. (1) **Clean/correlated cues, $kv \leq 64$: the gate reverses Hopfield** (0.99–1.00 vs 0.97–0.98, $\Delta + 0.02$, $n = 8$); at $\rho = 0.9$ the reversal *widens* ($\Delta + 0.08$ to $+0.16$) **only because Hopfield collapses on correlated keys** (0.72/0.59) while the CLU gate *also* drops (0.87/0.67) — Hopfield fragility, not CLU strength. (2) **Beyond kv64 is an epoch-budget wall, not a capacity wall**: at 2000 ep a wall is visible (gate 0.99 $\rightarrow$ 0.92 $\rightarrow$ 0.60 across kv64/96/128) but at 4000 ep kv96 reverses (0.975 vs 0.947, $\Delta + 0.03$) and kv128 only ties ($\Delta + 0.004$); required epochs scale with kv, and small cells **over-train** (kv32 gate 1.00 $\rightarrow$ 0.89, 2000 $\rightarrow$ 4000 ep). (3) **THE NOISE WALL (dominant negative, foregrounded)**: under cue noise $\sigma \in \{0.3, 0.6\}$ **no cell closes at any capacity, even kv32** (gate 0.36 vs Hopfield 0.71 at $\sigma=0.6$ /kv32; Δ from -0.05 to -0.35), *despite* CLU fidelity ≈ 1.0 — the relaxation gate is markedly less noise-robust than one matvec, on-thesis for the position (*the gate rations clean retrieval; noise-robustness is Hopfield’s*). **Tally at 2000 ep: 6/15 cells close** (all correlation-axis, $kv \leq 64$; eval-noise axis 0/6). Figure 2 is the honest regime map — (a) fidelity, (b) the clean-cue reversal, and (c) **the noise wall itself**, plotted rather than merely asserted; Figure 3 the epoch-scaling frontier; full tables in App. C.4. Net: **the gate reaches or exceeds Hopfield accuracy only for clean/correlated cues at $kv \leq 64$** (kv-scaled epochs extend this on the clean axis); Hopfield keeps both the cost and the cue-noise-robustness advantage; **the CLU asset is escalatable accuracy under a rationing gate on clean retrieval, not a general accuracy or cost win.** Separately, the anchor training intervention that rescues a designed vacuum does *not* transfer to memory fidelity (it pins a structureless init; App. D N30) — longer training, not the anchor, closes the gap.

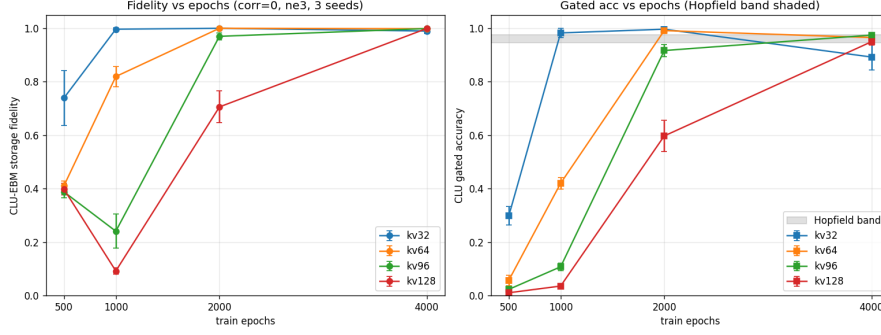


Figure 3: **Epoch-scaling frontier (§4.3, evidence).** Gate accuracy and storage fidelity vs training epochs across $kv \in \{32, 64, 96, 128\}$, against the epoch-independent Hopfield band (0.947–0.976): an epoch-budget ridge, not a capacity wall; kv32 over-trains ($1.00 \rightarrow 0.89$). (*Candidate for the appendix at the pruning pass.*)

5 Position, scope, and horizon

The position, restated. On a conservative memory, test-time compute is *paid access*, and the price list is physical: a squeeze buys escape for a bounded, re-absorbable injection and **prices reach exponentially** (it exceeds the box only by its instantaneous displacement $p_0 \sinh \zeta / M_0$, at energy cost $e^{2\zeta}$); a wormhole buys **unbounded** reach for a det $J = 1$ receipt and a *fixed* discrete ledger, and transports rather than destroys stored content — where an untrained det $J = 0$ **state-replacing** jump, which reaches by overwriting the state, provably erases it. (Deciding *whether* to take that certified edge is separate: §4.2’s learned decision head routes *through* the channel and inherits its receipt — decision and transport are orthogonal.) These are theorems verified to machine precision (§3), composing with the memory’s BIBO guarantees *only under coercive-component screening* — a bounded, or even free, energy ledger is not enough (App. B.2). Where a learned halting head or an energy verifier (Gladstone et al. 2025) escalates *the same operation* with no structural account, a symplectic memory prices each access mechanism *and certifies what it did to the state*. The honest pillars of §4 delimit where this buys a real ML advantage (escalatable rationing) and where a cheaper black box wins (routing): a certificate stack is a design discipline, not a guarantee of dominance.

Scope (stated, not buried). The §3 verifications use **oracle channel placement**, dim 2/4, 5 seeds, laptop-CPU; the §4 learned-memory pillars are MQAR-style, vocab-256, kv/ N small, laptop-CPU. No claim here is at scale, and none uses learned placement.

Horizon / future work.

- **Learned entrance-steering is the engineering crux.** The reach theorem proves the outer reachable set, but a trajectory must still *arrive* at a channel entrance under the flow; placing/learning (a, b) so entrances sit on natural trajectories is the true cost at scale and the likely failure point — addressed in forthcoming work.
- **Certifying exits on a learned potential.** The BIBO receipt (App. B.2) screens exits against the coercive component of an *analytic* well with closed-form x_b . For a learned, architecturally non-coercive V_θ the boundary is unknown; a sub-level-set estimator (or restoring

the $\alpha\|q\|^2$ confinement Deep/Conv drop) is the natural next experiment. Our coercive screen is an oracle, exactly as our channel placement is.

- **γ -re-absorption timing.** The governor re-absorption certificate ($t_{\text{reabsorb}} \approx 2\zeta/\gamma_c$) is verified only to leading order at $\gamma = 0$; a $\gamma > 0$ sweep closes that row.
- **Retry acceptance as a certified kernel** is developed as the closing design-rules below (derivations in App. F).

Test-time retries as a certified Markov kernel — four design rules (theory-complete on toy EBMs; no runs on trained CLU checkpoints are claimed). The paid-access logic extends one level up, from a single access mechanism to the *acceptance rule* governing repeated retries. Because the squeeze $S_\zeta^{(M)}$ is exactly symplectic ($\det J = 1$), a sign-symmetrized squeeze accepted by $\min(1, e^{-\Delta H/T})$ needs **no Jacobian correction** and is a π -reversible kernel for the Gibbs measure of the trained energy — test-time compute as MCMC with a *stationarity certificate* (Duane et al. 1987; Neal 2011). Four rules follow, each with its receipt, and each a restriction rather than a promise.

1. **Run certified segments at $\gamma = 0$; keep the governor outside them.** State-dependent $\gamma(H) > 0$ is non- π -preserving dissipation, so the actual squeeze-then-relax cascade is **Metropolis-within-annealing** toward a colder, MAP-seeking measure ($T_{\text{eff}} : 1.0 \rightarrow 0.61$ as $\gamma : 0 \rightarrow 0.2$, verified) — possibly *desirable for retrieval*, but **not Gibbs sampling**. **We never claim stationarity for the governed composite.**
2. **Mix; do not rely on the squeeze alone.** The squeeze family is a one-parameter subgroup whose orbit is a single hyperbola, hence **reducible/non-ergodic**; read it as a mass-metric-preconditioned *reach* move and use $\frac{1}{2}$ MALA(σ^*) + $\frac{1}{2}$ sign-symmetrized squeeze-MH (a mixture of π -reversible kernels is π -reversible).
3. **Calibrate the Langevin noise by fluctuation–dissipation (Newtonian kinetic mode).** The discrete FDT scale $\sigma_i^* = \sqrt{M_{\text{eff},i} T \gamma (2 - \gamma)}$ is load-bearing, not cosmetic: it moves the sampler from shadow-biased to exact (L_1 0.0995 $\rightarrow$ 0.0065, float64 toy EBMs). *Scope (a scope clause, not a caveat — V1’s units are Newtonian throughout):* this exactness is a Newtonian-mode statement. Once the step is Metropolis-adjusted, σ^* is a *proposal-tuning* scale (mixing efficiency), not a correctness condition — any σ leaves π invariant under MH, and the 0.0995 $\rightarrow$ 0.0065 gain is the finite-budget shadow bias it removes. In a relativistic kinetic mode the momentum target is Maxwell–Jüttner, for which a fixed-covariance Gaussian kick is not a Gibbs refresh, so there the MH correction — not σ^* — secures stationarity (App. F.2).
4. **Project every proposal off the coset tangent — even the certified retry carries a receipt.** The Gibbs measure is *exactly flat* along the Goldstone coset, so coset-tangent proposals are accepted w.p. 1 and **random-walk the stored register** at $D = \frac{1}{2}s^2$, erasing it after $N_{\text{erode}} \approx (\Delta_{\text{read}}/s)^2$ accepted moves [proven; verified $D = 1.29 \times 10^{-3}$ vs $\frac{1}{2}s^2 = 1.25 \times 10^{-3}$] — true even for a *charge*-preserving isotropic squeeze, which conserves Q while eroding coset *position*. Projection quenches it ($D = 0$, verified).

Per the energy-is-not-a-signal discipline held throughout: the acceptance temperature that makes the kernel *useful* is a per-model, write-time-calibrated quantity — the same learned gate §4.1 already fits — so **the value of the MH framing is the stationarity certificate and the explicit erosion accounting, not parameter parsimony, and not, on current evidence,**

performance. Derivations, the four numerical checks, and the specified discriminating experiment are in App. F.

Prior-art honesty. The nonlocal edge overlaps attention, skip connections, and MoE routing (Shazeer et al. 2017; Raposo et al. 2024); the squeeze overlaps simulated-annealing / basin-hopping / MCMC proposals; the certified-kernel reading sits in the Hamiltonian-Monte-Carlo (Duane et al. 1987; Neal 2011) and Metropolis-adjusted-Langevin (Roberts & Tweedie 1996) lineage (the squeeze a linear, gradient-free, mass-preconditioned replacement for the leapfrog trajectory; σ^* an FDT calibration, not a new algorithm); the coset-erosion result is Mermin–Wagner-flavoured; and the relativistic cap has a physics precedent in Lieb–Robinson light-cone bounds (Lieb & Robinson 1972). We cite these for the bound and lineage and claim only the *design consequences* — the $\det J = 1 + \text{energy-ledger} + \text{latch-transport certificate}$, and the governor-composed, coset-projected, FDT-calibrated, bounded-injection acceptance with an explicit certified-retry erosion budget — not novelty of nonlocality, stochastic escape, MCMC itself, or the causal bound.

A Flag-provenance tables

All results inherit their exact non-default configuration; see `draft.md` Appendix A for the full four tables (§3 certificate stack, commit 6f2384c; §4.1 gate, commit 572c708, with the memory-agnostic Hopfield transfer $0.18 \rightarrow 0.88$ from `minus-the-physics` Part B, 5 seeds; §4.2 routing, commit 52330f8; §4.3 regime map, commit 63fea62, `regime-remap-2000ep` 198 jobs, `train_epochs` $\in \{500, 1000, 2000, 4000\}$ the swept arm, $n=8$ capacity / $n=5$ stress / $n=3$ frontier; **§3.2.1 latch payoff + App. B.2 BIBO battery**, commit 27f232f (`v1-certificate-payoff`, JAX 0.9.0, no training, oracle placement, 5 seeds, dim 2 & 4)). Key non-defaults: §3 mass band $[4.0, 0.25]$, relativistic + Newtonian-control modes, $\gamma = 0$ reach rollout, $dt = 0.05$, $T = 100 \Rightarrow L = 2.5$; App. B.2 BIBO uses $\gamma = 0.02$, $k = 1.0$, $\epsilon = 0.02 \Rightarrow x_b = 3.5355$, $V_b = 3.125$, $T = 2000$ (r^* also at $2T$), escape radius 20.0; §4 relativistic kinetic, mlp potential, PCD write, `langevin_noise=legacy`. Cross-section note (two axes: damping and epochs). *Damping*: §3’s reach rollout uses $\gamma = 0$ for a sharp box while App. B.2 uses $\gamma = 0.02$ *because a bounded arm must be able to settle* (at $\gamma = 0$ a conservative orbit never converges, so “bounded” would be untestable). *Epochs (the axis governing the §4.1↔§4.3 comparison)*: **§4.1 trains 400 ep; §4.3 sweeps $\{500, 1000, 2000, 4000\}$ ep** — different budgets, reconciled inline in §4.1 point 2 (§4.1’s escalatable-accuracy headroom is a 400-ep, not-yet-converged property; §4.3 App. C.4.a gives the converged counterpart, where accuracy *matches* full-budget and the payoff is rationing only).

B The paid-access certificate table, and the two fine-print experiments

B.1 Table 1 — the receipt per mechanism

mechanism	energy injected (bound)	volume $\det J$	governor absorption	re-latch impact
wormhole (translation)	$\Delta V = V(b) - V(a)$, discrete, ledgered	$\mathbf{1}$ exactly; $(1-\gamma)^d$ damped	$\approx (1/\gamma_c) \ln(1+\Delta V/E^*)$	transport , shift $p^\top X \Delta$
wormhole (throat in V_θ)	standing $\sim A$, no jump	$(1-\gamma)^d$ exact	continuous	adds $\mu^2 \propto A/\ell^2$
squeeze $S^{(M)}$	$\leq (e^{2 \zeta }-1)H$ (matched quad. H)	$\mathbf{1}$ exactly	$\approx 2\zeta/\gamma_c$	preserved (off-coset)

BIBO caveat (with the table). Both compose with a coercive V_θ and $\gamma > 0$ (both $\det J \leq 1$). But Deep/Conv potentials are non-coercive out-of-unit; a wormhole exit into a non-coercive region can escape to infinity. **Design constraint:** exits inside a coercive sub-level *component* (or with $\alpha\|q\|^2$ confinement). This is now **exercised and measured** (B.2); the §3.2 reach testbed uses coercive wells, where the constraint is inactive.

B.2 The BIBO battery — a free ledger does not buy boundedness

Reporting grade: verification (designed analytic testbed; oracle placement). Source **v1-certificate-payoff** §7.4. We make the architectural non-coercivity of Deep/Conv potentials analytic and controllable with $V = \frac{1}{2}kq_0^2 - \epsilon q_0^4 + \frac{1}{2} \text{conf} \|q_1\|^2$ ($k = 1$, $\epsilon = 0.02$, $\text{conf} = 4.0$), coercive *only* inside the connected component $|q_0| < x_b = \sqrt{k/4\epsilon} = 3.536$; barrier $V_b = 3.125$. Three arms, six exits b ; $\gamma = 0.02$, $T = 2000$, relativistic, 5 seeds; $r^* = \max_t \|q_t\|$ measured at T and $2T$.

arm (r^* at $2T$)	$b=1.0$	2.0	3.0	3.6	4.0	5.0
wormhole + receipt (screened)	1.01	2.01	3.01	0.09	0.09	0.09
wormhole, receipt ignored (ablation)	1.01	2.01	3.01	102.13	103.43	104.83
state-replacing map (no receipt)	1.01	2.01	3.01	102.13	103.43	104.83
escape rate, certified	0	0	0	0	0	0
escape rate, blind / router	0	0	0	1.0	1.0	1.0
wormhole receipt	ADMIT	ADMIT	ADMIT	REJECT	REJECT	REJECT
<i>energy-only sub-level test</i>	admit	admit	admit	reject	admit	admit

(T -values for the escaping arms: 52.12/53.43/54.83.)

- The diagnostic is a growth rate, not a big number.** $r^*(2T)/r^*(T) = 1.000$ for every bounded arm/exit (saturated) vs 1.96/1.94/1.91 for the escaping ones — $r^* \propto T$, genuinely unbounded, terminal velocity capped at $c/\sqrt{M_0} = 0.5$ by the relativistic governor. **The causal cap bounds speed, not excursion.**
- The receipt predicts BIBO blow-up on 6/6 exits;** the ledger $\Delta H = V(b) - V(a)$ is exact to 0.0 at every exit.

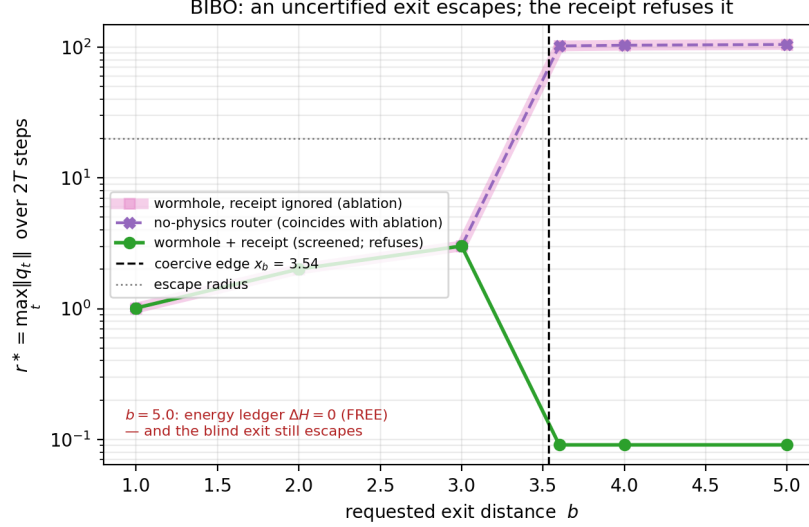


Figure 4: **BIBO battery (App. B.2, verification grade)**. An uncertified exit escapes ($r^* \propto T$); the receipt-screened exit stays bounded by *refusing* to jump. The router curve coincides *exactly* with the receipt-ignored wormhole ablation (drawn as a translucent band under a dashed overlay so neither is occluded) — the honest reading is that the *receipt*, not the jump, buys boundedness. Note $b = 5.0$: the energy ledger is free ($\Delta H = 0$) and the exit still escapes.

3. **The killer row is $b = 5.0$.** $V(5.0) = 0.0$ *exactly*: the ledger says the jump is **free** — cheaper than the admissible $b = 3.0$ exit ($\Delta H = 2.88$) — and an energy-only sub-level test **admits** it. It escapes anyway; likewise $b = 4.0$ ($V = 2.88 < V_b$; note $V(3.0) = V(4.0) = 2.88$ exactly, a coincidence of $\frac{1}{2}x^2 - 0.02x^4$). $\det J = 1$ **plus a bounded — or even zero — ledger is NOT sufficient for BIBO; coercive-component membership is the operative clause.**
4. **Attribution, honestly.** wormhole_blind and no_physics_router coincide exactly: neither is screened. So what buys BIBO is **the receipt, not the jump mechanism**. The defensible claim: *the wormhole can form the receipt* (it has V_θ , an exact ledger, a well-defined Jacobian) *and a physics-free state-replacing map has none of the three and cannot screen its own exit even in principle* — bolting a coercivity check onto it means handing it the certificate machinery. **Payoff A (§3.2.1) carries no such caveat, which is why it, not this, is the headline panel.**
5. **The trade the certificate makes explicit.** On rejected exits the certified wormhole **does not reach the target** ($r^* = 0.09$; it stays home). The receipt does not make an unsafe exit safe — it *refuses* it.
6. **Scope.** The coercive screen uses the analytic x_b : an **oracle** too. Certifying exits on a trained, non-coercive V_θ is genuinely open. Also $b = 3.6$ sits 0.06 outside x_b , yet $r^*(2T) = 102$ vs 0.09 and growth 1.96 make the classification unambiguous — no knife-edge, unlike the §3.2 squeeze bracket.

C Full honest tables

See `draft.md` Appendix C for: (C.1) the full reach battery grid + per-arm certificates (wormhole $\det J = [1.0] \times 6$ exact, ledger err $[0.0] \times 6$; **state-replacing map (no_physics_router)** $\det J = [0.0] \times 6$, **measured by forward-mode autodiff — volume-annihilating and non-invertible, not merely uncertified**; squeeze $\det S^{(M)} = 1.000 \pm 4e-6$; forbidden state-dependent gate $\det J = 2.05$) and (C.1.b) the latch-payoff cloud for all four arms with replication across $\dim \in \{2, 4\} \times \text{seed} \in \{0, 7\}$; (C.2) the routing grid ($N \in \{4, 8\}$, all arms, FLOP/query — router $1.000/0.948 @ 8.81e7$ vs gated $0.887/0.715 @ 1.18e8$; chain $1.76e8 \rightarrow 2.94e8$); (C.3) Hopfield iteration parity (ceiling in 1 matvec at $\beta \geq 5$); (C.4) the settled full-grid regime map — (C.4.a) paired capacity axis 500 vs 2000 ep ($n=8$: kv32/kv64 reverse $\Delta + 0.02$, kv96 $\Delta - 0.04$), (C.4.b) epoch-scaling frontier (4000 ep closes kv96 $+0.03$, kv128 ties; kv32 over-trains $1.00 \rightarrow 0.89$), (C.4.c) stress axes including **the noise wall** (eval-noise 0/6 close: gate 0.36 vs Hopfield 0.71 at $\sigma=0.6/\text{kv32}$ despite fidelity ≈ 1.0) and the $\rho = 0.9$ Hopfield-collapse artifact. Tally 6/15 close at 2000 ep.

D Negatives on the record

Tier-A V1 negatives (future-work anchors, not hidden failures): **N1** squeeze retries recover nothing at single-unit scale — *tested selection among reachable, not crossing to unreachable*; does not bear on §3 (§3.3). **N2** abstention-vs-Hopfield unwinnable; Hopfield dominant at 500 ep (*provisional — under-trained map*). **N3** CLU energy $\approx$ readout margin ($\Delta \text{AUROC} \in [-0.004, +0.024]$) — *the forbidden energy-superiority claim, not made*. **N24** energy-gated wormhole loses to a physics-free router (the §4.2 boundary). **N30** $V(\text{data})$ -anchor does not transfer to memory fidelity. **N31** state-dependent jump-gate breaks volume by $\nabla g \cdot \Delta$ ($\det J = 2.05$; the frozen-gate design guard). **N23** smooth z -gate mis-routes $\sim 10\text{--}12\%$. Scope caveat: N24/N27 rest on a linearly-separable-cue laptop testbed — honest for the tested regime, not proven universal.

Fine print of Payoff A — volume preservation alone is *not* the latch receipt (measured design guard). The random-shift arm is symplectic ($\det J = 1$) and *still* scrambles the Goldstone charge (ΔQ std 0.2465; out-spread $0.2793 = 3.48 \times$ the incoming spread). Only the **matched channel** (Δ coset-tangent, $X\Delta \perp p$) makes ΔQ constant and preserves the spread. Any reading of §3.2.1 as “ $\det J = 1 \Rightarrow$ memory is safe” is wrong, and we flag it so the fine print cannot be inverted into the review.

Fine print of Payoff B — the receipt buys BIBO, *not* the jump; and the state-replacing map $\equiv$ a blind wormhole. The `wormhole_blind` and `no_physics_router` (state-replacing) arms coincide exactly on all six exits, so no boundedness advantage attaches to the wormhole’s jump mechanism per se — only to the screening receipt, which a physics-free state-replacing map cannot form (no V_θ , no ledger, no Jacobian). And a **free** ledger ($\Delta H = 0$ at $b = 5.0$) is admitted by an energy-only test and escapes anyway. This is why the latch (Payoff A), not BIBO (Payoff B), is the headline panel.

Payoff scope. Both payoffs are on the designed testbed: oracle channel placement, analytic potentials, no training. They demonstrate the certificate’s *downstream consequence*; they do **not** claim a real-data or learned-memory win. The coercive screen itself uses the analytic x_b and is likewise an oracle.

E Analytic verifications

`checks.py`, numpy float64, seed 0; double well $V = \beta(q^2 - a^2)^2$, $a = \beta = 1$, $\varepsilon = 0.05$. (A) reach cap $|\Delta q| = T\varepsilon c/\sqrt{M}$ to rel. err 2.0e-12; (B) plain trapped $q = -1.068$, wormhole $q = +1.000$, ledger 0.0; (C) $\det J$ const = 1.000, state-dep matches $1 + \nabla g \cdot \Delta$ to 2.2e-16; (D) squeeze hop threshold $\zeta^* = 0.2356$ vs measured 0.27; (E) latch transit err 1.8e-15, $X\Delta \perp p$ shift 0.0; (F) re-absorption 56 vs predicted 49.5 steps, injection $E_{\text{inj}}/E^* = 2.718 = e^{2 \cdot 0.5}$.

F Test-time retries as a certified Markov kernel (derivations, checks, specified experiment)

Source: `thread9-mh-kernel` (4 numerical checks, float64, toy EBMs). **Grade: theory-complete on toy EBMs. No runs on trained CLU checkpoints are claimed anywhere in this paper.** This appendix carries the derivation behind the four design rules of §5.

F.1 Why the squeeze needs no Jacobian correction. $S_\zeta^{(M)}$ is exactly volume-preserving ($\det J = 1$) — the property HMC (Duane et al. 1987; Neal 2011) exploits to avoid a Jacobian correction in the acceptance ratio. A retry proposing a **sign-symmetrized** squeeze ($\pm\zeta$ with equal probability, restoring proposal symmetry) and accepting with $\min(1, e^{-\Delta H/T})$ is a **detailed-balance kernel for $e^{-H/T}$** — test-time compute as MCMC with a stationarity certificate. π -reversibility is proven and verified ($L_1 = 0.0095$ between forward and reversed two-step histograms). The certificate is **narrower than the retry-cascade pitch assumes**; three physics facts sharpen it.

F.2 (i) Squeeze-only is non-ergodic; MALA supplies mixing, FDT supplies its scale. $\{S_\zeta\}$ is a one-parameter subgroup whose orbit of any point is a **single hyperbola** in (q_i, p_i) : squeeze-only proposals are **reducible/non-ergodic** and cannot reach off-hyperbola states at all. Ergodicity requires HMC-style momentum refreshment; the squeeze is best read as a **mass-metric-preconditioned global (reach) move** layered on a MALA step (Roberts & Tweedie 1996). Its noise scale is not free: the discrete fluctuation–dissipation relation for the damped Verlet step fixes $\sigma_i^* = \sqrt{M_{\text{eff},i} T \gamma (2 - \gamma)}$, and this is **load-bearing rather than cosmetic** — the FDT-correct σ^* moves the sampler from shadow-biased to exact, L_1 error $0.0995 \rightarrow 0.0065$ (15 $\times$; float64). **Kinetic-mode scope (Newtonian only; a scope clause, not a retraction — V1’s units are Newtonian):** (a) once Metropolis-adjusted, σ^* is a proposal-tuning scale (mixing efficiency), not a correctness condition — any σ leaves π invariant, and the $0.0995 \rightarrow 0.0065$ gain is the finite-budget shadow bias removed; (b) the Gaussian FDT kick is a Gibbs momentum-refresh only in the Newtonian modes, where π_p is Gaussian — in a relativistic mode π_p is Maxwell–Jüttner, a fixed-covariance Gaussian refresh is not one of its refreshes, and stationarity is secured by the MH correction rather than by any σ^* . A mixture of two π -reversible kernels is π -reversible, so $\frac{1}{2}\text{MALA}(\sigma^*) + \frac{1}{2}\text{squeeze-MH}$ inherits the certificate.

F.3 (ii) The governor destroys the invariant: the real cascade is annealing, not sampling. State-dependent $\gamma(H) > 0$ is non- π -preserving dissipation, so the deployed squeeze-then-relax cascade is **Metropolis-within-annealing**, converging to a colder MAP-seeking measure: $T_{\text{eff}} : 1.0 \rightarrow 0.61$ as $\gamma : 0 \rightarrow 0.2$ (verified, float64). This may be *desirable for retrieval* — a MAP-seeking cascade is what an associative memory wants — but **it is not Gibbs sampling, and we never claim stationarity for the governed composite**. Hence design rule 1.

F.4 (iii) Decisive for a memory: even the certified retry erodes the latch. The Gibbs measure is **exactly flat along the Goldstone coset** (no restoring force along a broken continuous symmetry), so any coset-tangent proposal has $\Delta H = 0$ and is accepted w.p. 1. Accepted coset-tangent moves compose into an unbiased random walk that **diffuses the stored register** at $D = \frac{1}{2}s^2$ per accepted move, with $N_{\text{erode}} \approx (\Delta_{\text{read}}/s)^2$, where s is the proposal scale and Δ_{read} the read-out margin along the coset [proven; verified $D = 1.29 \times 10^{-3}$ vs $\frac{1}{2}s^2 = 1.25 \times 10^{-3}$]. This holds **even for a charge-preserving isotropic squeeze**, which conserves $Q = p^\top X q$ while eroding coset *position* — charge and position are different observables and only the former is protected. Projecting proposals off the coset tangent quenches it ($D = 0$, verified), so only explicit writes move the register.

F.5 The deflation we state rather than bury. Acceptance is an *intra-model* quantity: cross-model incomparability of H does not break validity, but the operating temperature T that makes the kernel *useful* is model-specific and FDT-miscalibrated in general. A per-model T therefore **partially re-imports the learned gate** of §4.1 — the parsimony argument is weak. **The value of the MH framing is the stationarity certificate and the explicit erosion accounting, not parameter parsimony, and not, on current evidence, performance.**

F.6 The specified discriminating experiment (future work). Three arms (certified mixture kernel at $\gamma = 0$ with coset projection; the learned- τ escalation gate of §4.1; naive squeeze retries) on a trained CLU associative memory, matched on FLOPs, with **the latch-erosion decay curve as the money plot** — predicting that naive certified retries erode the register at $D = \frac{1}{2}s^2$ while the coset-projected kernel holds it flat. Risks: (a) the erosion rate is derived on toy EBMs and a trained- V_θ coset is only approximately flat; (b) the FDT σ^* assumes the damped-Verlet noise model, which `langevin_noise=legacy` runs violate; (c) matched-FLOPs accounting must charge the MALA gradient. **Not run; no CLU-checkpoint result is claimed.**

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